

IML Assignment-1

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1 Relationship Between Separate and Joint Linear Regressions

We are given monthly observations

$$(x_i, y_{i1}, y_{i2}), \quad i = 1, \dots, n$$

where

$$x_i \in \mathbb{R}$$

denotes the number of marriages, while

$$y_{i1}, y_{i2} \in \mathbb{R}$$

represent the weight of flowers sold and the weight of gold purchased respectively.

Define the response vector

$$y_i = \begin{bmatrix} y_{i1} \\ y_{i2} \end{bmatrix}$$

and define the matrices

$$X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \quad Y = \begin{bmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \\ \vdots & \vdots \\ y_{n1} & y_{n2} \end{bmatrix}.$$

Separate Linear Regressions

The flower sellers fit the regression model

$$y_{i1} = \theta_F x_i + \theta'_F$$

by minimizing the squared error

$$L_F(\theta_F, \theta'_F) = \sum_{i=1}^n (y_{i1} - \theta_F x_i - \theta'_F)^2.$$

The normal equation solution is

$$\theta_F = (X^T X)^{-1} X^T y_1$$

where

$$y_1 = \begin{bmatrix} y_{11} \\ y_{21} \\ \vdots \\ y_{n1} \end{bmatrix}.$$

Similarly, the jewellers fit

$$y_{i2} = \theta_J x_i + \theta'_J$$

with objective

$$L_J(\theta_J, \theta'_J) = \sum_{i=1}^n (y_{i2} - \theta_J x_i - \theta'_J)^2.$$

The corresponding normal equation solution is

$$\theta_J = (X^T X)^{-1} X^T y_2$$

where

$$y_2 = \begin{bmatrix} y_{12} \\ y_{22} \\ \vdots \\ y_{n2} \end{bmatrix}.$$

The two groups combine their estimates into the matrix

$$\Theta_{FJ} = \begin{bmatrix} \theta_F & \theta_J \end{bmatrix}.$$

Joint Multivariate Regression

With full access to the data, we instead model

$$y_i = \Theta^T x_i + \Theta'$$

and minimize the squared error

$$L(\Theta) = \sum_{i=1}^n \|y_i - \Theta^T x_i - \Theta'\|^2.$$

In matrix form this becomes

$$L(\Theta) = \|Y - X\Theta\|_F^2$$

where $\|\cdot\|_F$ denotes the norm.

The normal equation solution is

$$\Theta = (X^T X)^{-1} X^T Y.$$

Writing Y column-wise,

$$Y = \begin{bmatrix} y_1 & y_2 \end{bmatrix},$$

we obtain

$$\Theta = (X^T X)^{-1} X^T \begin{bmatrix} y_1 & y_2 \end{bmatrix}.$$

Matrix multiplication distributes across columns:

$$\Theta = \begin{bmatrix} (X^T X)^{-1} X^T y_1 & (X^T X)^{-1} X^T y_2 \end{bmatrix}.$$

Substituting the expressions for the separate regressions,

$$(X^T X)^{-1} X^T y_1 = \theta_F, \quad (X^T X)^{-1} X^T y_2 = \theta_J,$$

we obtain

$$\Theta = \begin{bmatrix} \theta_F & \theta_J \end{bmatrix}.$$

Thus

$$\Theta = \Theta_{FJ}.$$

Therefore, solving the joint multivariate regression is equivalent to solving the two separate regressions and stacking the resulting coefficient vectors.

Gradient Descent Interpretation

Consider the joint loss

$$L(\Theta) = \sum_{i=1}^n \|y_i - \Theta^T x_i\|^2.$$

Expanding the squared norm,

$$\|y_i - \Theta^T x_i\|^2 = (y_{i1} - \theta_F x_i)^2 + (y_{i2} - \theta_J x_i)^2.$$

Thus the total loss decomposes as

$$L(\Theta) = \sum_{i=1}^n (y_{i1} - \theta_F x_i)^2 + \sum_{i=1}^n (y_{i2} - \theta_J x_i)^2.$$

Hence

$$L(\Theta) = L_F(\theta_F) + L_J(\theta_J).$$

Gradient descent updates take the form

$$\Theta^{(t+1)} = \Theta^{(t)} - \eta \nabla L(\Theta^{(t)}).$$

Because the loss separates into two independent components, the gradients satisfy

$$\nabla_{\theta_F} L = \nabla L_F, \quad \nabla_{\theta_J} L = \nabla L_J.$$

Thus the updates become

$$\theta_F^{(t+1)} = \theta_F^{(t)} - \eta \nabla L_F(\theta_F^{(t)}),$$

$$\theta_J^{(t+1)} = \theta_J^{(t)} - \eta \nabla L_J(\theta_J^{(t)}).$$

Therefore gradient descent on the joint matrix parameter is equivalent to running the two gradient descent algorithms independently and stacking the parameters at each iteration:

$$\Theta^{(t)} = \begin{bmatrix} \theta_F^{(t)} & \theta_J^{(t)} \end{bmatrix}.$$

2 Kernel Regression

2.1 (a) Polynomial Kernel

We are given the function

$$K(x, y) = (x^\top y + c)^l, \quad l \geq 1.$$

A function $K(x, y)$ is a valid kernel if there exists a feature map $\phi(x)$ such that

$$K(x, y) = \phi(x)^\top \phi(y).$$

Using the binomial theorem,

$$(x^\top y + c)^l = \sum_{k=0}^l \binom{l}{k} (x^\top y)^k c^{l-k}.$$

Now note that

$$(x^\top y)^k = \left(\sum_{i=1}^d x_i y_i \right)^k.$$

Expanding this expression produces all monomials of degree k :

$$(x^\top y)^k = \sum_{|\alpha|=k} \frac{k!}{\alpha_1! \cdots \alpha_d!} x^\alpha y^\alpha$$

where

$$x^\alpha = x_1^{\alpha_1} \cdots x_d^{\alpha_d}, \quad |\alpha| = \alpha_1 + \cdots + \alpha_d.$$

Substituting this expansion into the kernel expression yields

$$K(x, y) = \sum_{k=0}^l \binom{l}{k} c^{l-k} \sum_{|\alpha|=k} \frac{k!}{\alpha_1! \cdots \alpha_d!} x^\alpha y^\alpha.$$

Define the feature map

$$\phi(x) = \left[\sqrt{\binom{l}{k} c^{l-k} \frac{k!}{\alpha_1! \cdots \alpha_d!}} x^\alpha \right]_{|\alpha| \leq l}.$$

Then

$$\phi(x)^\top \phi(y) = (x^\top y + c)^l.$$

Therefore $K(x, y)$ is a valid kernel.

2.2 (b) A Non-valid Kernel

Consider

$$K(x, y) = (x^\top y)^2 - x^\top y.$$

If K were a valid kernel then the Gram matrix

$$K_{ij} = K(x_i, x_j)$$

must be positive semidefinite for every dataset.

Take

$$x_1 = 1, \quad x_2 = 2.$$

Then

$$K(x_1, x_1) = 1^2 - 1 = 0,$$

$$K(x_1, x_2) = (2)^2 - 2 = 2,$$

$$K(x_2, x_2) = (4)^2 - 4 = 12.$$

Thus the Gram matrix is

$$K = \begin{bmatrix} 0 & 2 \\ 2 & 12 \end{bmatrix}.$$

A valid kernel must produce a positive semidefinite Gram matrix. However this matrix has a negative eigenvalue and therefore is not positive semidefinite.

Hence $K(x, y)$ is not a valid kernel.

2.3 (c) Gaussian Kernel

We now prove that

$$K(x, y) = \exp\left(-\frac{\|x - y\|^2}{2\sigma^2}\right)$$

is a valid kernel.

First expand the squared distance:

$$\|x - y\|^2 = \|x\|^2 + \|y\|^2 - 2x^\top y.$$

Substituting this into the kernel gives

$$K(x, y) = \exp\left(-\frac{\|x\|^2}{2\sigma^2}\right) \exp\left(\frac{x^\top y}{\sigma^2}\right) \exp\left(-\frac{\|y\|^2}{2\sigma^2}\right).$$

Define

$$f(x) = \exp\left(-\frac{\|x\|^2}{2\sigma^2}\right).$$

Then

$$K(x, y) = f(x) \exp\left(\frac{x^\top y}{\sigma^2}\right) f(y).$$

Closure properties If K_1 and K_2 are kernels with feature maps ϕ_1 and ϕ_2 , then

$$K_1(x, y) + K_2(x, y)$$

is also a kernel with feature map

$$\phi(x) = \begin{bmatrix} \phi_1(x) \\ \phi_2(x) \end{bmatrix}.$$

Similarly, for $c > 0$,

$$cK_1(x, y)$$

is a kernel with feature map $\sqrt{c}\phi_1(x)$.

Product of kernels If

$$K_1(x, y) = \phi_1(x)^\top \phi_1(y), \quad K_2(x, y) = \phi_2(x)^\top \phi_2(y),$$

then the Gram matrices of K_1 and K_2 are positive semidefinite. The Gram matrix of the product kernel

$$K(x, y) = K_1(x, y)K_2(x, y)$$

is the Hadamard product of these matrices. By the Schur product theorem, the Hadamard product of positive semidefinite matrices is positive semidefinite. Hence the product is also a kernel.

Exponential of a kernel Using the Taylor expansion

$$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!},$$

if $K(x, y)$ is a kernel then each power $K(x, y)^n$ is also a kernel, and therefore

$$\exp(K(x, y)) = \sum_{n=0}^{\infty} \frac{K(x, y)^n}{n!}$$

is a kernel as a positive weighted sum of kernels.

Applying this to the linear kernel $x^\top y$ shows that

$$\exp\left(\frac{x^\top y}{\sigma^2}\right)$$

is a kernel.

Multiplying this kernel by $f(x)f(y)$ preserves validity. Therefore

$$K(x, y) = \exp\left(-\frac{\|x - y\|^2}{2\sigma^2}\right)$$

is a valid kernel.

3 Logistic Regression Model

We are given observations (x_i, y_i) where

$$x_i \in \mathbb{R}$$

denotes the age at which a person was fired and

$$y_i \in \{0, 1\}$$

indicates whether the person remained unemployed for more than one year.

Model Definition

Let

$$p(x) = \Pr(Y = 1 \mid X = x)$$

denote the probability that a person fired at age x remains unemployed for more than one year.

The *odds* of the event are defined as

$$\text{odds}(x) = \frac{p(x)}{1 - p(x)}.$$

Logistic regression assumes that the log-odds are linear in the predictor. Thus there exist parameters θ_0, θ_1 such that

$$\log\left(\frac{p(x)}{1-p(x)}\right) = \theta_0 + \theta_1 x.$$

Exponentiating both sides gives

$$\frac{p(x)}{1-p(x)} = e^{\theta_0 + \theta_1 x}.$$

Thus the odds function is

$$\text{odds}(x) = e^{\theta_0 + \theta_1 x}.$$

First Observation: Change in Odds

We are told that if the firing age increases by one year, the odds increase by 1.75%.

An increase of 1.75% corresponds to multiplying the odds by

$$1 + \frac{1.75}{100}.$$

Thus

$$\text{odds}(x+1) = \left(1 + \frac{1.75}{100}\right) \text{odds}(x).$$

Substituting the logistic form,

$$e^{\theta_0 + \theta_1(x+1)} = \left(1 + \frac{1.75}{100}\right) e^{\theta_0 + \theta_1 x}.$$

Dividing both sides by $e^{\theta_0 + \theta_1 x}$ gives

$$e^{\theta_1} = 1 + \frac{1.75}{100}.$$

Taking logarithms,

$$\theta_1 = \log\left(1 + \frac{1.75}{100}\right).$$

Second Observation: Odds at Age 34

We are given that

$$\text{odds}(34) = 0.5.$$

Using the logistic model,

$$e^{\theta_0 + 34\theta_1} = 0.5.$$

Taking logarithms,

$$\theta_0 + 34\theta_1 = \log(0.5).$$

Solving for the Parameters

Substituting the expression for θ_1 gives

$$\theta_0 = \log(0.5) - 34 \log\left(1 + \frac{1.75}{100}\right).$$

Final Parameter Expressions

$$\theta_1 = \log\left(1 + \frac{1.75}{100}\right)$$

$$\theta_0 = \log(0.5) - 34 \log\left(1 + \frac{1.75}{100}\right)$$

4 Decision Tree Construction

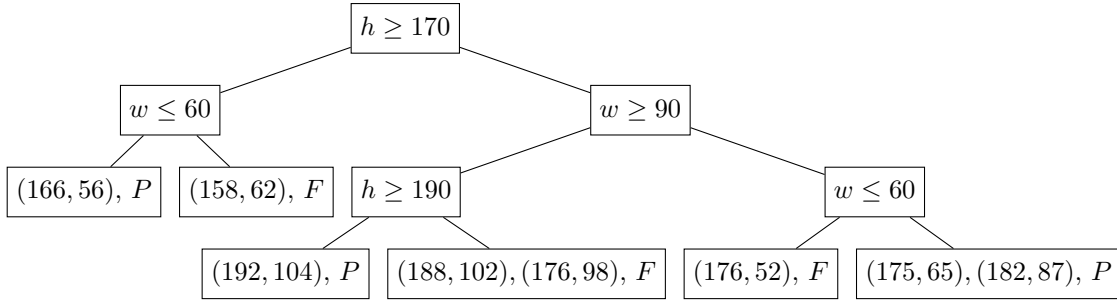
We construct a decision tree that fits the dataset with zero classification error. The splits are restricted to axis-aligned rules on height or weight. We begin with the proposed splits

$$h \geq 170 \quad \text{followed by} \quad w \geq 90.$$

The dataset is

$$(175, 65, P), (158, 62, F), (188, 102, F), (176, 98, F),$$

$$(182, 87, P), (176, 52, F), (166, 56, P), (192, 104, P).$$



5 Naive Bayes Spam Classification

The dataset provides the number of spam and non-spam messages containing each word.

word	spam	non-spam
win	20	15
free	40	15
money	30	15
meeting	0	60
this	30	45
is	20	30

Total Counts

$$N_S = 20 + 40 + 30 + 0 + 30 + 20 = 140$$

$$N_H = 15 + 15 + 15 + 60 + 45 + 30 = 180$$

Thus the conditional probabilities are

$$P(w|S) = \frac{\text{spam count of } w}{140}, \quad P(w|H) = \frac{\text{non-spam count of } w}{180}.$$

Using the naive Bayes assumption,

$$P(S|w_1, \dots, w_k) \propto P(S) \prod_i P(w_i|S),$$

$$P(H|w_1, \dots, w_k) \propto P(H) \prod_i P(w_i|H).$$

Since the class priors are not given, we assume

$$P(S) = P(H),$$

so the priors cancel when comparing the likelihoods.

Email 1: “This is free money meeting”

Words:

{this, is, free, money, meeting}

Spam likelihood:

$$P(\text{email}|S) = \frac{30}{140} \cdot \frac{20}{140} \cdot \frac{40}{140} \cdot \frac{30}{140} \cdot \frac{0}{140} = 0$$

Non-spam likelihood:

$$P(\text{email}|H) = \frac{45}{180} \cdot \frac{30}{180} \cdot \frac{15}{180} \cdot \frac{15}{180} \cdot \frac{60}{180}$$

Since

$$P(\text{email}|S) = 0$$

the message is classified as **non-spam**.

Email 2: “Free this, win money”

Words:

{free, this, win, money}

Spam likelihood:

$$P(\text{email}|S) = \frac{40}{140} \cdot \frac{30}{140} \cdot \frac{20}{140} \cdot \frac{30}{140}$$

Non-spam likelihood:

$$P(\text{email}|H) = \frac{15}{180} \cdot \frac{45}{180} \cdot \frac{15}{180} \cdot \frac{15}{180}$$

Since the spam likelihood is larger, the message is classified as **spam**.

Email 3: “Meeting is free”

Words:

{meeting, is, free}

Spam likelihood:

$$P(\text{email}|S) = \frac{0}{140} \cdot \frac{20}{140} \cdot \frac{40}{140} = 0$$

Non-spam likelihood:

$$P(\text{email}|H) = \frac{60}{180} \cdot \frac{30}{180} \cdot \frac{15}{180}$$

Since

$$P(\text{email}|S) = 0$$

the message is classified as **non-spam**.

References

- Class notes
- <https://www.ethanepperly.com/index.php/2025/02/25/the-schur-product-theorem/> for Schur Product
- Andrew Ng notes
- Wikipedia pages for kernel methods, decision trees, Naive Bayes
- ChatGPT was used to convert my handwritten solution to latex code.
- DeepSeek was used to help me understand approaches to solve some problems